

# Draft Note on Mixed-State Interface Structure in a Lindbladian Chern-Insulator Workspace

Paper-style synthesis from code, cached data, and rendered figures

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## Abstract

This note summarizes a numerical workspace devoted to Lindbladian evolution of a two-dimensional free-fermion Chern-insulator-like model in a domain-wall geometry. The main evidence points to a persistent interface sector in the mixed-state covariance matrix: spatially localized wall modes,  $k_y$ -resolved branches traversing between near-empty and near-filled occupation sectors, a strip-like local Chern-marker pattern, and wall-centered current and entanglement signatures. The conceptual background for dissipative Chern preparation and mixed-state topology is already well established in the literature. The narrower possible contribution suggested here is the combination of an *internal* domain wall with a multi-diagnostic study of mixed-state interface phenomenology. Because the present repository shows some code-version drift, the note is written as a careful draft rather than a final claim of reproducible topological classification.

## 1 Introduction

Engineered dissipation can stabilize nontrivial steady states of quadratic fermionic systems, including topological ones [1, 2, 7]. At the same time, the mixed-state character of Lindbladian steady states forces some care in the use of topological language: several diagnostics familiar from pure-state band theory must be reinterpreted in terms of density matrices, covariance matrices, or purity spectra [3, 4, 5].

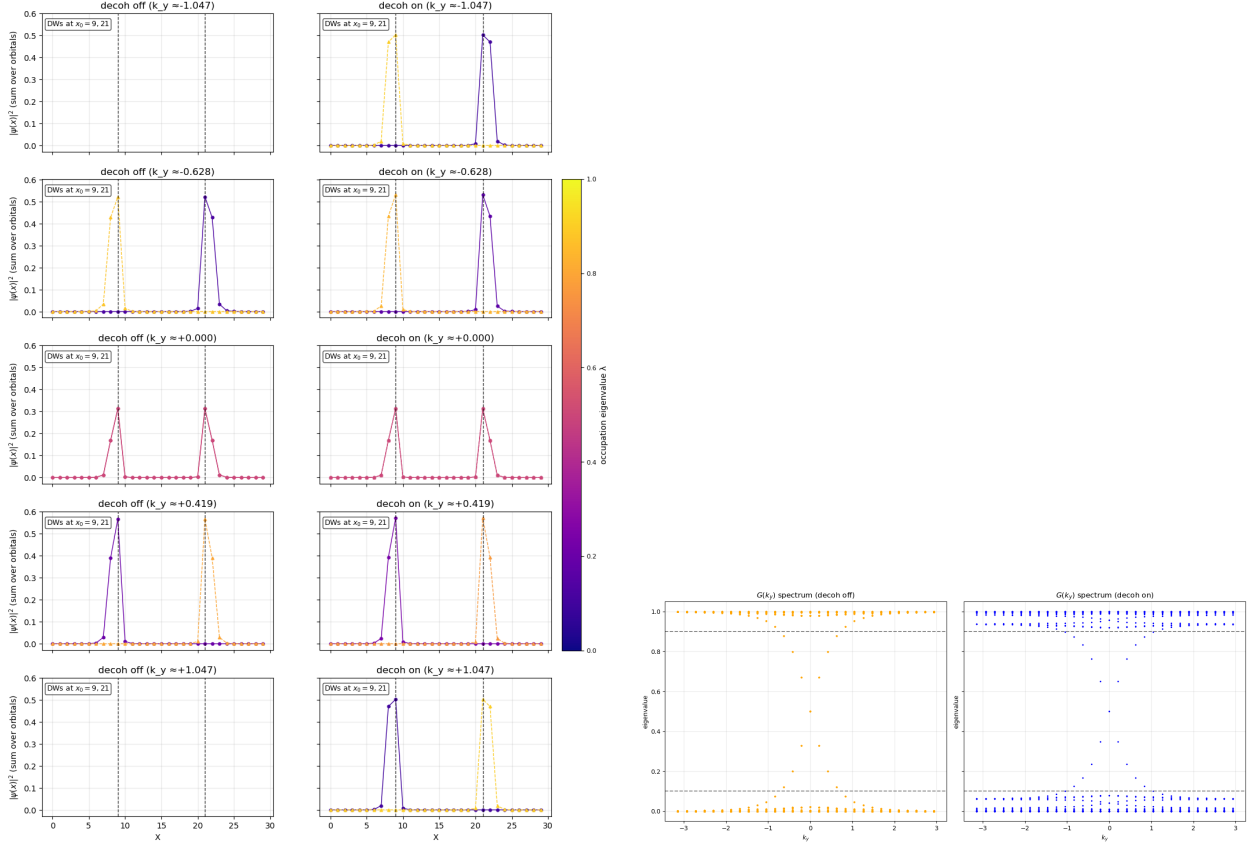
The present workspace appears to explore precisely this regime for a Chern-insulator-like free-fermion model with a deliberately introduced domain wall. The central question suggested by the code and figures is not whether dissipative Chern preparation is possible in principle; that question has prior answers. Rather, the interesting question is whether a mixed dissipative state retains a distinguished *interface sector* when a topological strip is embedded inside a trivial background, and whether that sector can be tracked consistently across spatial, spectral, correlation, current, and entanglement-like observables.

## 2 Model and Diagnostics

The core object in the repository is a single-particle correlation matrix  $G$  evolving under a quadratic Lindbladian. In the uniform sector, the code uses a two-band Chern-insulator-like Bloch structure

$$n_x(\mathbf{k}) = \sin k_x, \quad n_y(\mathbf{k}) = \sin k_y, \quad n_z(\mathbf{k}) = \alpha - \cos k_x - \cos k_y, \quad (1)$$

from which band projectors and overcomplete Wannier objects are constructed in `CI_Lindblad.py`. The domain-wall extension in `CI_Lindblad_DW.py` replaces the uniform mass by a strip geometry with two interfaces. In the current code defaults, this is most naturally read as a topological strip



(a) Domain-wall-localized mode weight for representative  $k_y$  values, with the interface positions visibly selected.

(b)  $k_y$ -resolved covariance-matrix spectrum showing branches that pass between near-empty and near-filled occupation sectors.

Figure 1: Primary evidence for a mixed-state interface sector.

with  $\alpha_2 = 1$  embedded in a trivial background with  $\alpha_1 = 3$ . For  $N_x = 31$ , the current implementation places the two walls at  $x = 9$  and  $x = 21$ .

The diagnostics stored in the repository are unusually broad for a single workspace. They include local Chern-marker maps,  $k_y$ -resolved spectra, time-resolved spectra, spatial correlation profiles, entanglement-contour plots, perturbative response runs, and current-like maps. This breadth matters: the physical picture is not inferred from one plot alone, but from the consistency of several observables with the same interface-centered narrative.

### 3 Numerical Phenomenology

#### 3.1 Interface localization and mixed-state spectral flow

The most immediate evidence for interface physics is shown in Fig. 1. The left panel displays covariance-matrix eigenstate weight sharply concentrated near two spatial locations coinciding with the domain-wall positions. The right panel shows  $k_y$ -resolved branches that interpolate between occupation sectors near 0 and near 1. In a purely Hamiltonian context, one would describe this as edge or interface spectral flow. Here the object is a mixed-state covariance spectrum, so the language should remain more cautious, but the visual structure is nonetheless edge- or wall-like.

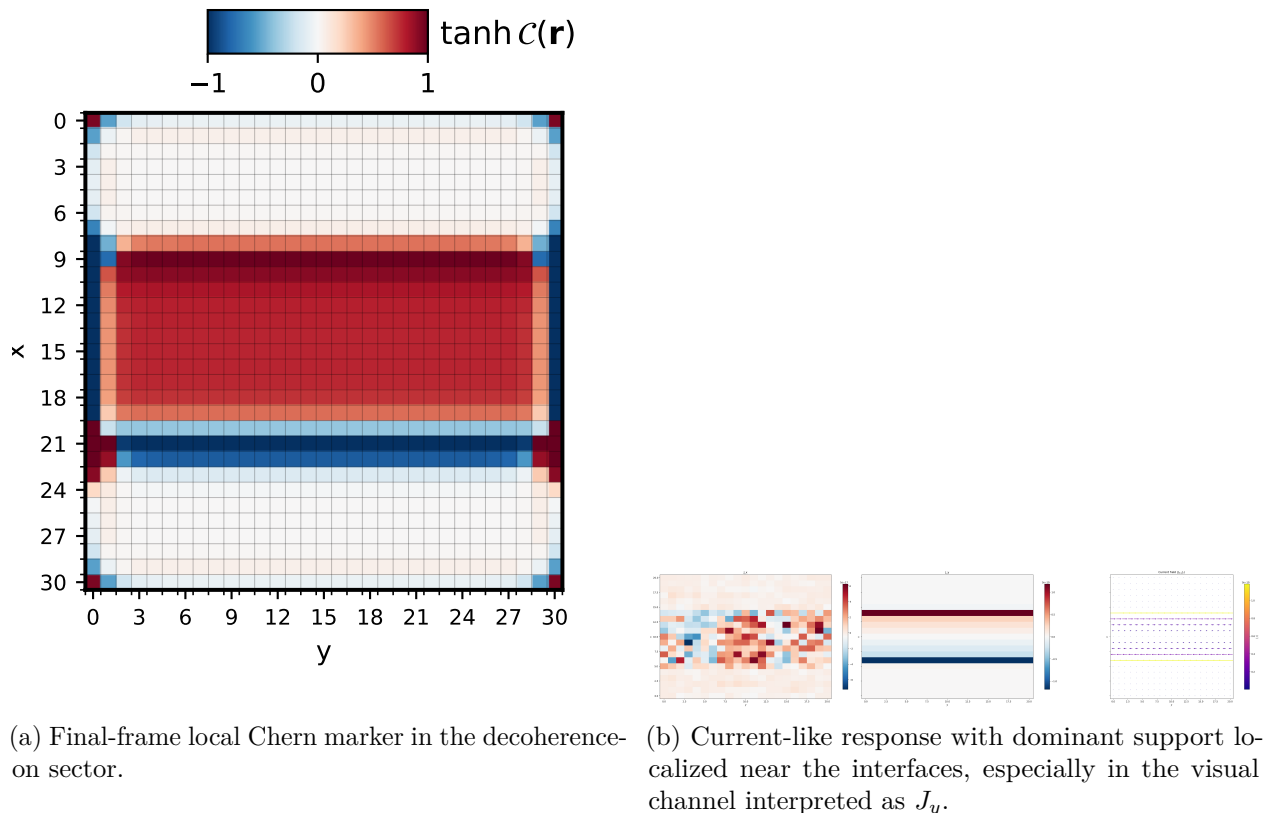


Figure 2: Real-space diagnostics consistent with an interface-centered steady sector.

### 3.2 Real-space marker structure and current response

The real-space marker plots reinforce the same story. The local Chern marker is not featureless in the late-time state; instead, it organizes into a strip-like pattern that tracks the nonuniform geometry. This is precisely the type of spatial organization one would expect if the interface retains a topological or quasi-topological identity in the mixed steady state. The current-like maps support the same reading, showing a wall-centered response rather than a uniform bulk pattern.

### 3.3 What can and cannot yet be claimed

At the present stage, the repository supports a strong *phenomenological* statement: the domain wall is the most structured part of the late-time mixed state, and several diagnostics agree on that point. What is not yet established is the strongest possible interpretation of this pattern. In particular, the current evidence does not by itself prove a rigorously quantized mixed-state topological invariant, nor does it yet distinguish cleanly between a protected chiral interface mode and a numerically robust dissipative midgap/localized mode. That distinction likely requires a tighter reproducibility pass and more formal analysis of the mixed-state invariant being diagnosed.

## 4 Relation to Existing Literature

The broad theme of the workspace sits inside a mature literature. Budich, Zoller, and Diehl gave a direct baseline for dissipative preparation of two-dimensional Chern-insulator states [1].

More general open-system topological structure via engineered dissipation was developed earlier in one-dimensional wire settings by Diehl *et al.* [2]. Mixed-state topology has been framed by Budich and Diehl [3], while Bardyn *et al.* and Wawer and Fleischhauer developed probes and invariants for Gaussian mixed states [4, 5]. The local Chern-marker background comes from Bianco and Resta [6]. Finally, Goldstein emphasized an important limitation: in dimensions above one, finite-range Lindbladians do not generically yield finite-rate relaxation into a unique topological *pure* state, though mixed states arbitrarily close to the target one remain possible [7].

There is also relevant work closer to the geometry of the present figures. Shavit and Goldstein studied transport in dissipatively induced Chern states [8]. Yang, Mognini, and Bergholtz analyzed dissipative boundary-state preparation, including a two-dimensional Chern-insulator setting [9]. On the coherent or transport side, Upadhyaya and Tserkovnyak, and later Rosen *et al.*, studied chiral transport along domain walls in quantum anomalous Hall systems [10, 11].

The right novelty claim is therefore narrow. Figure 1 is *not* broadly novel if taken to mean simply “domain walls host localized and spectrally distinct modes” or “topological/dissipative Chern systems can show boundary-state phenomenology.” Those ideas already have clear precedent. What may still be novel is the combination realized by this workspace: an *internal* domain wall inside a Gaussian Lindbladian Chern-insulator workflow, with a mixedness-sensitive interface sector diagnosed simultaneously by localization, covariance spectra, local marker structure, correlations, currents, and entanglement-like observables.

## 5 Discussion

The main strength of the repository is that the interface story is overdetermined: it is visible in more than one observable family. This makes the domain-wall interpretation substantially more convincing than if it relied on a single spectral plot. The main weakness is reproducibility: some saved artifacts carry labels such as `alpha_triv30` that do not map transparently onto the current constructor surface, which suggests code-version drift between figure generation and the present source tree.

For that reason, the most responsible conclusion is a conditional one. If the cached figures can be tied cleanly to the current scripts, and if the mixed-state diagnostics can be sharpened enough to distinguish topology from generic interface localization, then this workspace could support a worthwhile short paper on dissipative domain-wall phenomenology in a mixed Gaussian Chern-insulator setting. Without that tightening, the figures remain interesting, but their conceptual status should stay provisional.

## 6 Conclusion

This workspace already contains a coherent paper-sized narrative. A quadratic Lindbladian Chern-insulator model with an internal domain wall appears to develop a persistent mixed-state interface sector whose signatures are visible in real space and in momentum-resolved spectra. The main opportunity is not to claim a new general principle of dissipative topology, but to make a carefully delimited case for domain-wall-centered mixed-state interface physics inside that broader framework.

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